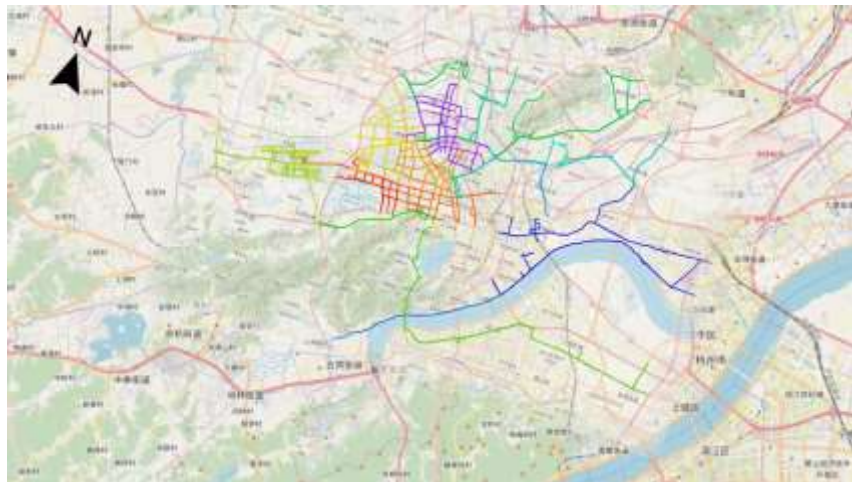


Introduction

- Multimodal Computer Vision: Introduction to Vision and Multimodal Learning
- Zhiwen Fan (zhiwenfan@tamu.edu)
- Research Interests:
 - Computer vision, machine learning
 - 3D reconstruction and understanding from images
 - Generative models for reasoning and content creation

Zhiwen's work

- 3D reconstruction, visualization, simulation



City-scale reconstruction



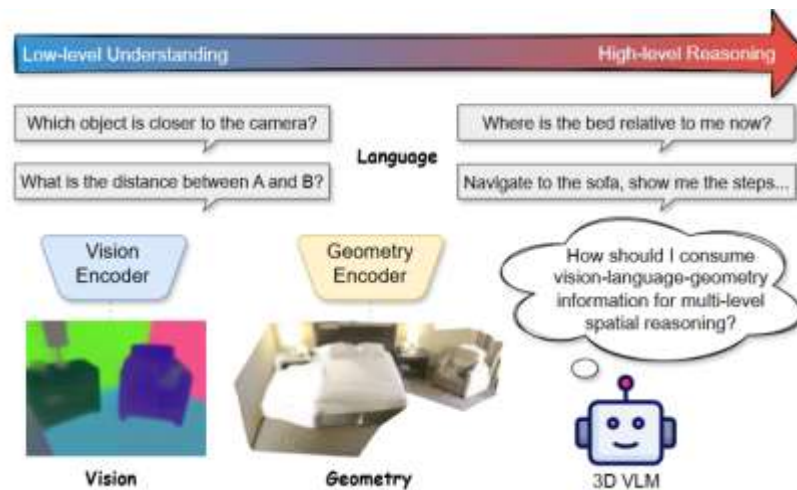
Indoor scene reconstruction

Zhiwen's work

- 3D generation and reasoning



VR/AR under Quest 3



Embodied reasoning

Today

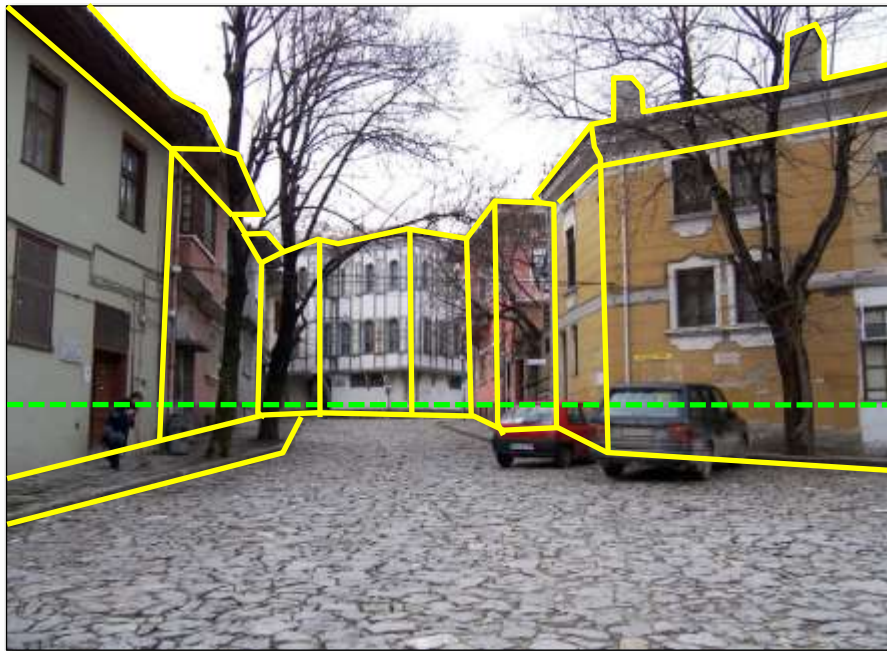
- Introduction to Multimodal Computer Vision

A picture is worth a thousand words



RGB Image.

A picture is worth a thousand words



Geometry understanding.

A picture is worth a thousand words



Semantic understanding.

A picture is worth a thousand words



please describe this image.

ChatGPT 5.2 Thinking ▾



The image shows a quiet, old-looking street paved with uneven cobblestones. Bare trees line both sides, suggesting winter or late fall. On the right is a yellow, weathered building with peeling patches, and two cars are parked along the curb (a red car and a dark gray one). On the left are older residential buildings, and a person is walking near the edge of the street. In the distance, the road curves toward a larger white building with tall windows under an overcast sky.

A picture is worth a thousand words



Affordance.

A picture is worth a thousand words



Actionable information.

A picture is worth a thousand words

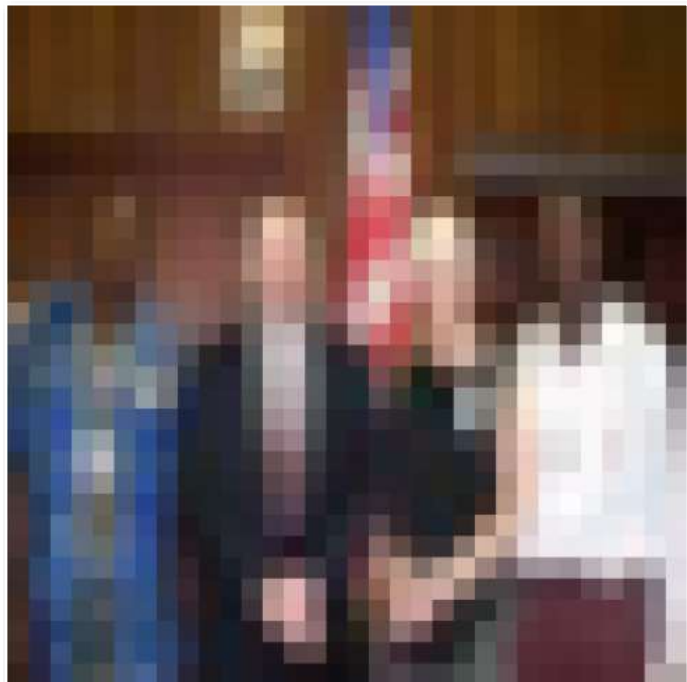
please generate an image, that shows a quiet, old-looking street paved with uneven cobblestones. Bare trees line both sides, suggesting winter or late fall. On the right is a yellow, weathered building with peeling patches, and two cars are parked along the curb (a red car and a dark gray one). On the left are older residential buildings, and a person is walking near the edge of the street. In the distance, the road curves toward a larger white building with tall windows under an overcast sky.

ChatGPT 5.2 Thinking ▾



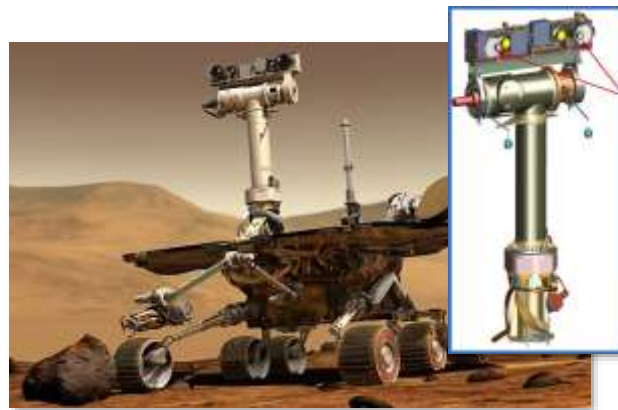
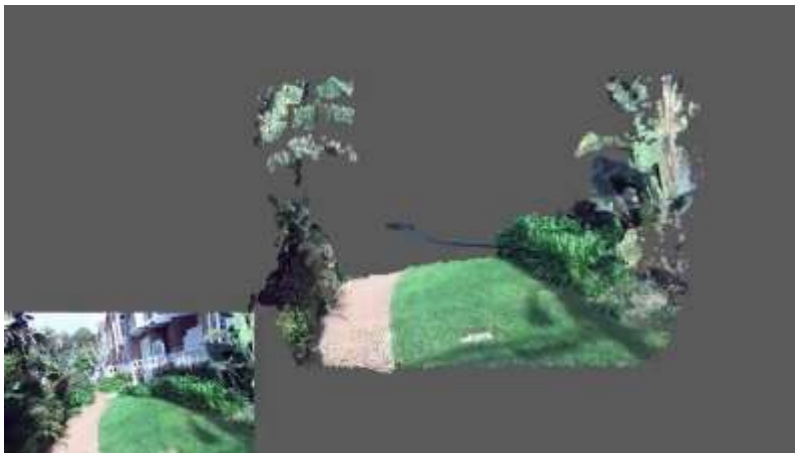
Image generation.

Humans are quite good at image understanding



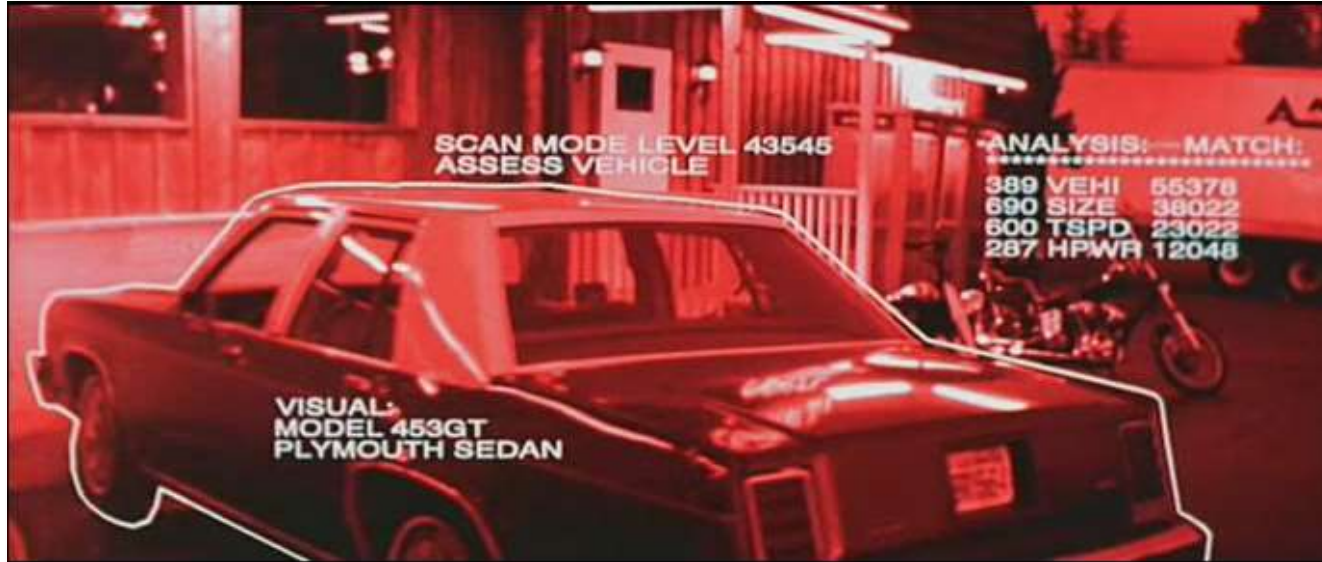
A group photo with ambiguity.

The goal of computer vision



Compute the 3D shape of the world

The goal of computer vision



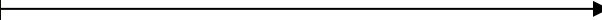
Recognize objects and people from sensor reading

The goal of computer vision



Image segmentation

The goal of computer vision



Enhance image quality.

The goal of computer vision



Super-resolution (source: 2d3)



Low-light photography

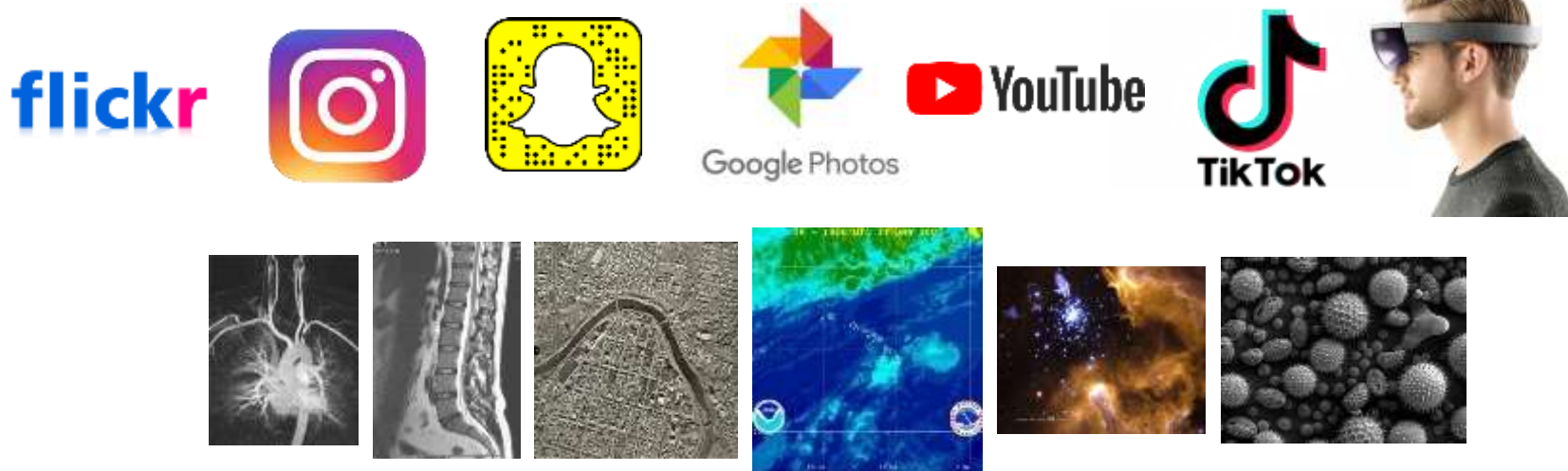
Enhance image quality.



Removing objects

Why study computer vision?

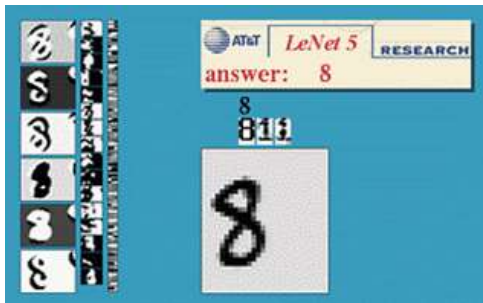
- Billions of images/videos captured per day



- Huge number of potential applications

Applications: Optical character recognition (OCR)

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs (1990's)
<http://yann.lecun.com/exdb/lenet/>



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



Automatic check processing



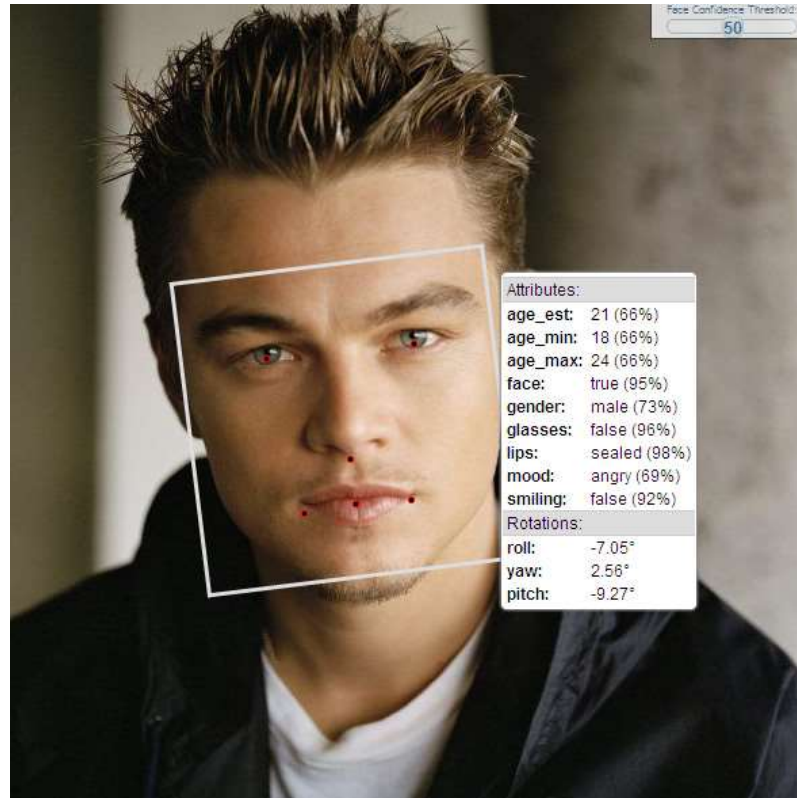
Sudoku grabber
<http://sudokugrab.blogspot.com/>

Applications: Face detection

- Nearly all cameras detect faces in real time (Why?)



Applications: Face analysis and recognition



Applications: FaceID login without a password

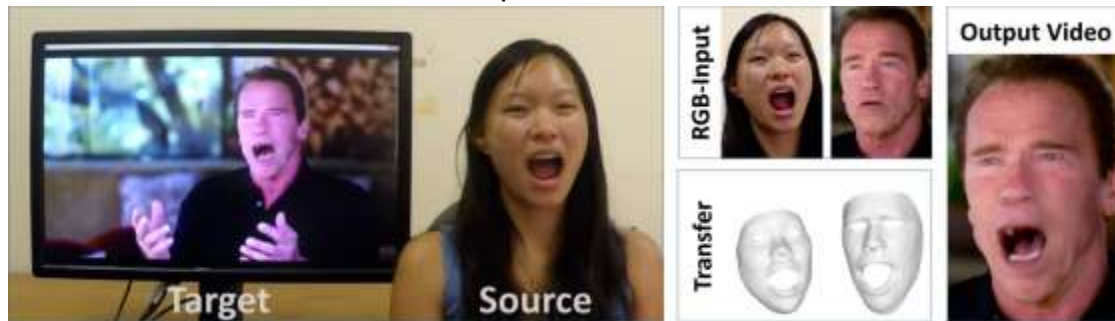


Face unlock on Apple iPhone X
See also <http://www.sensiblevision.com/>

Applications: 3D face tracking



Snapchat Lenses



Face2Face system (Thies et al.)

Applications: Image synthesis



Karras, et al., *Progressive Growing of GANs for Improved Quality, Stability, and Variation*, ICLR 2018

Applications: Image synthesis



“An astronaut riding a horse in a photorealistic style” – DALL-E 2



“A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat” – Imagen

Applications: Smart cars

- Tesla FSD, NVIDIA Alphamayo, Waymo

manufacturer products consumer products

Our Vision. Your Safety.

rear looking camera forward looking camera side looking camera

> **EyeQ** Vision on a Chip

> **Vision Applications**
Road, Vehicle, Pedestrian Protection and more

> **AWS** Advance Warning System

News

- > Hologeye advances Technologies Power Volvo Cars World First Collision Warning With Auto Brake System
- > Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end

Events

- > Hologeye at Equip Auto, Paris, France
- > Hologeye at SEMA, Las Vegas, NV

> read more

Applications: Robotics



NASA's Mars Curiosity Rover



Amazon Picking Challenge

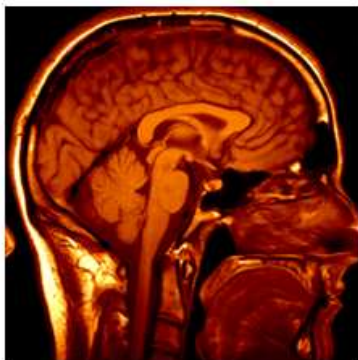


Amazon Prime Air

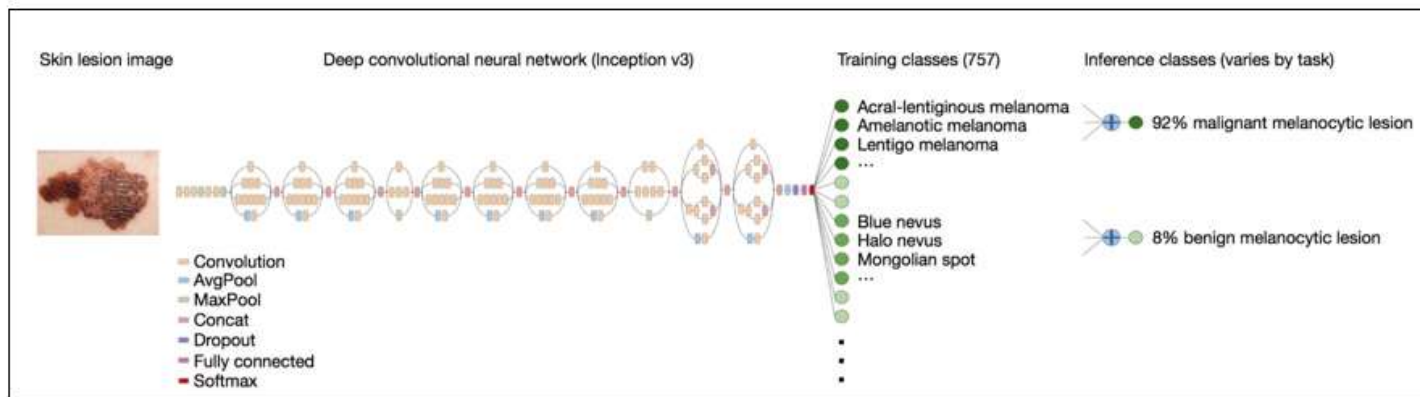


Amazon Scout

Applications: Medical imaging



3D imaging
(MRI, CT)



Skin cancer classification with deep learning
<https://cs.stanford.edu/people/esteva/nature/>

Applications: VR/AR industry

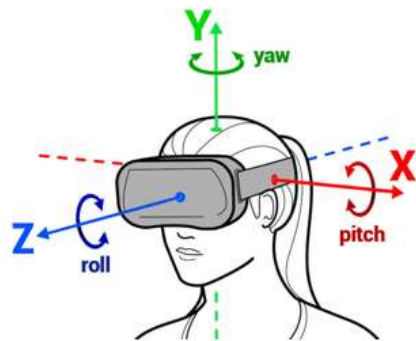
INVESTING 3/25/2014 @ 5:43PM | 70,399 views

Facebook Buys Oculus, Virtual Reality Gaming Startup, For \$2 Billion

[+ Comment Now](#) [+ Follow Comments](#)



Applications: VR/AR industry



6DoF head tracking



Hand & body tracking



3D scene understanding



3D-360 video capture

Current state of the art

You just saw many examples of current systems.

- Many of these are less than 5 years old

Computer vision is an active research area, and rapidly changing

- Many new apps in the next 5 years
- Deep learning and generative methods powering many modern applications

Many startups across a dizzying array of areas

- Generative AI, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, ...

Why is computer vision difficult?



Viewpoint variation



Illumination



Credit: Flickr user michaelpaul

Scale

Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)

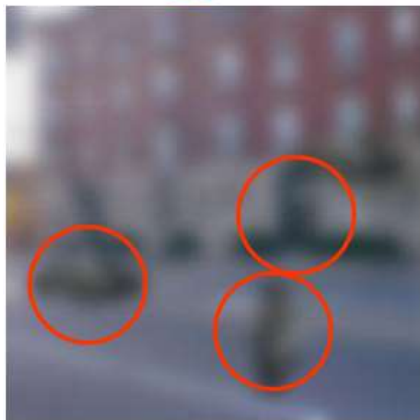


Background clutter



Occlusion

Why is computer vision difficult?



Why is computer vision difficult?

- Perception is an inherently ambiguous problem
 - Many different 3D scenes could have given rise to a given 2D image



- We often must use prior knowledge about the world's structure

ECEN 689: Multimodal Computer Vision

- Project-based course whose goal is to teach you the advanced progress of computer vision – image processing, geometry, recognition – in a hands-on way

ECEN 689: Multimodal Computer Vision

Week	Topic	Assignments
1 (2026/01/12)	Introduction to Vision and Multimodal Learning	
2 (2026/01/19)	Image Processing (classification, segmentation, detection); Paper presentation registration due	
3 (2026/01/26)	Neural Networks, and It's Training	
4 (2026/02/02)	Principles in 3D Vision: Multi-View Geometry; Paper presentation	- Assignment: run an existing NeRF or Gaussian-splatting demo; analyze quality vs. compute/time (coding + 1-page summary)
5 (2026/02/09)	Neural Radiance Fields and 3D Gaussian Splatting; Paper presentation	
6 (2026/02/16-)	3D Scene Understanding; Paper presentation	
7 (2026/02/23)	Real-time 3D Mapping; Paper presentation	Midterm Project release (individual work): survey 6–10 papers on your chosen topic (2+ pages, NeurIPS style)
8 (2026/03/02-)	Text to Image Generation; Paper presentation	
9 (2026/03/09-)	Text to 3D Generation; Paper presentation	- Midterm Literature Review Report Due
10 (2026/03/16)	Spring break, no class	
11 (2026/03/23)	Self-Supervised Pre-training; Paper presentation	
12 (2026/03/30)	Large Language Models and Vision-Language Models; Paper presentation	- Assignment: Identify and evaluate VLMs on tasks where they perform poorly, such as spatial navigation in complex 3D scenes. (coding + 1-page summary)
13 (2026/04/06)	Vision-Language Models for Autonomous Systems; Paper presentation	Final project release: team up with overall 4 students, improve the algorithm based on chosen topic and write report.
14 (2026/04/13)	Autonomous Driving; Paper presentation	
15 (2026/04/20)	Final Project Presentations	
16 (2026/04/27, last day of class)	Final Project Presentations	- Submit final project report (9 pages, NeurIPS-style)

ECEN 689: Multimodal Computer Vision

- Prerequisites
 - ECEN 447 or ECEN 642 (grade of C or better)
 - Data structures
 - Python coding
 - Linear algebra

ECEN 689: Multimodal Computer Vision

- Participation (10%)
Regular attendance, active engagement in discussions, and thoughtful questions.
- Two Assignment (20%)
Take-home research project: submit code samples and a one-page summary (NeurIPS style).
- Individual Paper Presentations (15%)
 - Each student presents one or two papers on a selected topic.
 - Graded on clarity, depth of understanding, and quality of communication.
 - Slides including background, problem definition, method, experiments, and future works.
- Midterm Literature Review Report (20%)
 - Expand upon a presented paper or chosen research direction by surveying 6–10 related papers.
 - Write a concise report (2+ pages, excluding references) using the NeurIPS 2025 LaTeX template.
 - The report should include:
 - ✓ Introduction and motivation
 - ✓ Background overview
 - ✓ Descriptions of key algorithms, their interconnections, and salient experimental findings
 - ✓ Clear organization and synthesis of insights
- Final Team Project (35%)
 - Final Project Presentation (15%)
 - ✓ Detailed presentation covering problem statement, methodology, experiments, results, and analysis.
 - ✓ Graded on clarity, depth of analysis, and quality of delivery.
 - Final Project Report (20%)
 - ✓ A written report (9+ pages, excluding references; supplementary material encouraged) in NeurIPS 2025 LaTeX format.
 - ✓ Should include: introduction, related work, approach, experimental design and results, discussion, and conclusion.
 - ✓ Aim for workshop-level quality at a top-tier venue.
 - ✓ Graded on novelty, completeness, rigor of experiments, clarity of writing, and demonstration of effort.

ECEN 689: Multimodal Computer Vision

- Late policy
 - Up to one late submission will be accepted.
 - A late assignment/project will be penalized by 10% for each day it is late, and no extra credit will be awarded

ECEN 689: Multimodal Computer Vision

- Lecture (Every Tuesday)
- Two assignment
 - Coding + 1-page report
- Two paper presentations (Every Thursday)
 - Each week we have two presentation slots
 - Register the paper of related topics on the sheet
 - Prepare slides for presentation, from motivation to methods to results
- Midterm project
 - Individual small project: coding + 4-page report
- Final project
 - Group-wise project: coding + 9-page report